Project: Enterprise Network Segmentation & Gateway Routing

Architecture Overview

This project involved designing and implementing a segmented network architecture to facilitate secure communication between two distinct departments. By leveraging a **Cisco 2911 ISR**, I established a centralized routing gateway to bridge different logical subnets.

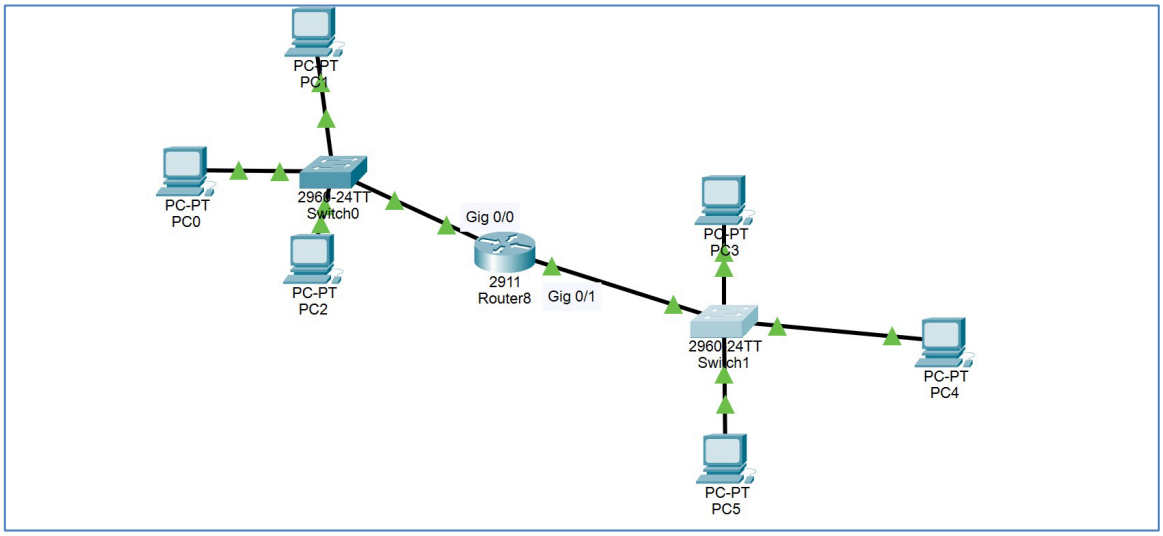
Network Topology

The infrastructure is divided into two primary segments to optimize traffic flow and management:

- **Internal Operations Segment:** A three-node workstation cluster (Subnet: 192.168.1.0/24) aggregated through a **Cisco 2960 Access Switch**.

- **Remote/Management Segment:** A second three-node cluster (Subnet: 172.16.100.0/24) connected via an independent **Cisco 2960 Access Switch**.

- **Inter-Subnet Routing:** Both access switches are uplinked to dedicated GigabitEthernet interfaces on the primary router, enabling controlled inter-VLAN traversal



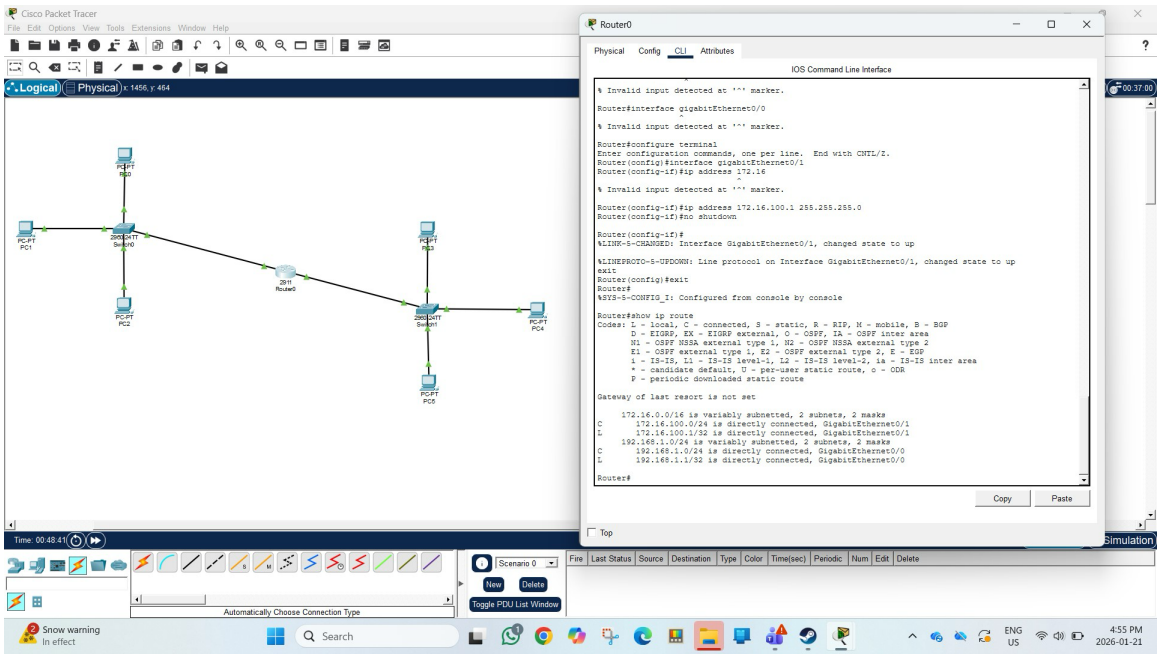
2. IP Addressing:

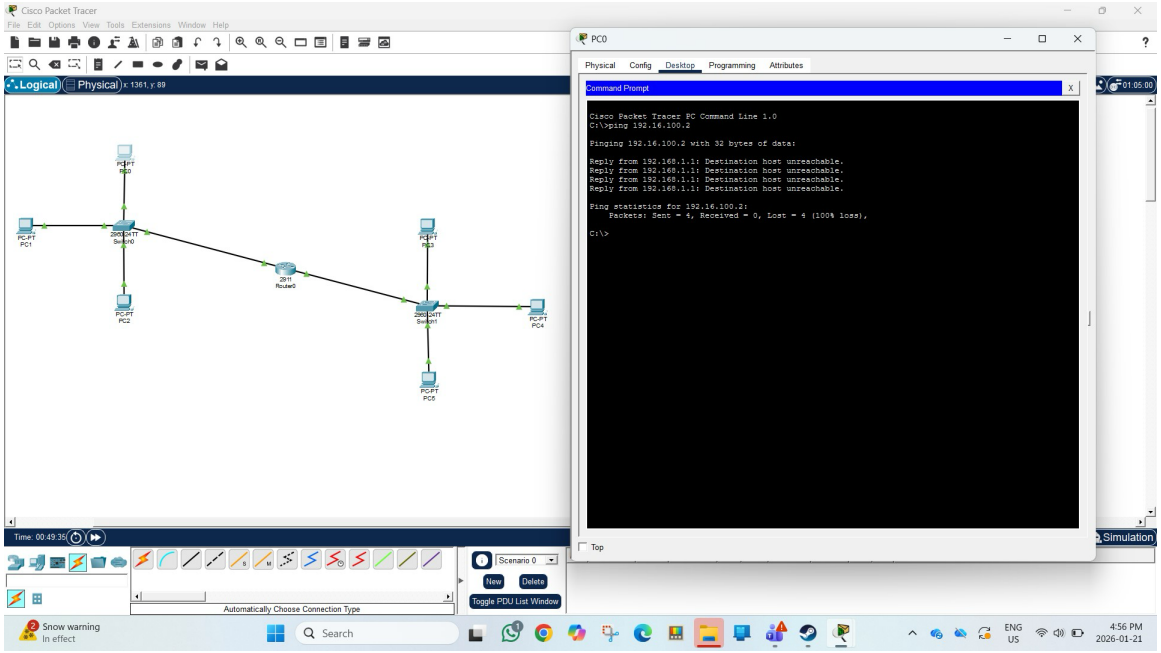
- Assign different subnets to the devices on either side of the router.
- Configure static routes in the router to allow communication between the subnets.

IP Address Plan

Device	Interface	IP Address	Subnet Mask	Subnet
PC0	Ethernet 0	192.168.1.2	255.255.255.0	192.168.1.0/24
PC1	Ethernet 0	192.168.1.3	255.255.255.0	192.168.1.0/24
PC2	Ethernet 0	192.168.1.4	255.255.255.0	192.168.1.0/24
Router (Switch0)	GigabitEthernet 0/0	192.168.1.1	255.255.255.0	192.168.1.0/24
PC3	Ethernet 0	172.16.100.2	255.255.255.0	172.16.100.0/24
PC4	Ethernet 0	172.16.100.3	255.255.255.0	172.16.100.0/24
PC5	Ethernet 0	172.16.100.4	255.255.255.0	172.16.100.0/24
Router (Switch1)	GigabitEthernet 0/1	172.16.100.1	255.255.255.0	172.16.100.0/24

Router Interface Provisioning:





Infrastructure Validation & Connectivity Audit

To ensure the integrity of the inter-segment routing, the following validation procedures were performed:

1. **End-to-End ICMP Reachability:** Verified successful packet traversal between the 192.168.1.0/24 (Operations) and 172.16.100.0/24 (Management) subnets. This confirms that the router is correctly processing cross-segment traffic and that the gateways are properly advertised to all endpoints.

2. **Routing Table Integrity Check:** Audited the Cisco IOS routing table using the show ip route command. This verified that the router has successfully established "Directly Connected" paths for both interfaces and is actively maintaining a Layer 3 map of the environment.